

/ Tests pytest

Optimized testing with different operating systems and python versions

```
- name: Login into the GitHub container registry
  uses: docker/login-action@f4ef78c080cd8ba55a85445d5
  with:
    password: ${ secrets.GITHUB_TOKEN }
    registry: ghcr.io
    username: ${ secrets.GH_USERNAME }

- env:
  LICENSE_SERVER: ${ secrets.LICENSE_SERVER }
  name: Pull, launch and verify desired Ansys service
  run: '...'

- name: Run pytest with desired markers and extra args
  uses: ansys/actions/tests-pytest@v10.3
  with:
    pytest-extra-args: ${ matrix.cfg.extra-args }
    pytest-markers: ${ matrix.cfg.markers }
    requires-xvfb: true
```

Ref: [Optimized testing with different operating systems and Python versions](#)

Testing library with different operating systems and python versions

```
- name: Run pytest
  uses: ansys/actions/tests-pytest@v10.3
  with:
    pytest-extra-args: --cov=ansys.<library> --cov-report=html
    pytest-markers: -k 'mocked'
```

Ref: [Testing library with different operating systems and Python versions](#)

/ Hk migrate fork pr

Migrate fork pr to main repository

```
- name: Migrate fork PR
  uses: ansys/actions/hk-migrate-fork-pr@v10.3
  with:
    bot-email: ${ secrets.PYANSYS_CI_BOT_EMAIL }
    bot-username: ${ secrets.PYANSYS_CI_BOT_USERNAME }
    comment-id: ${ github.event.comment.id }
    github-token: ${ secrets.PYANSYS_CI_BOT_TOKEN }
    pr-number: ${ github.event.issue.number }
    team-slug: pyansys-maintainers
    user-triggering: ${ github.event.comment.user.login }
```

Ref: [Migrate fork PR to main repository](#)

/ Check actions security

Check actions security

```
- uses: ansys/actions/check-actions-security@v10.3
  with:
    auditing-level: high
    generate-summary: true
    token: ${ secrets.GITHUB_TOKEN }
    trust-ansys-actions: true
```

Ref: [Check actions security](#)

/ Doc build

Installing additional system dependencies for building documentation

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@v10.3
  with:
    dependencies: graphviz mermaid-cli
```

Ref: [Installing additional system dependencies for building documentation](#)

Building library documentation that using xvfb

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@v10.3
  with:
    requires-xvfb: true
```

Ref: [Building library documentation that using XVFB](#)

Building library documentation

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@v10.3
```

Ref: [Building library documentation](#)

/ Doc deploy pr

Deploy pr documentation

```
- uses: ansys/actions/doc-deploy-pr@v10.3
  with:
    bot-email: ${ secrets.PYANSYS_CI_BOT_EMAIL }
    bot-user: ${ secrets.PYANSYS_CI_BOT_USERNAME }
    cname: ${ env.DOCUMENTATION_CNAME }
    maximum-pr-doc-deployments: 3
    token: ${ secrets.GITHUB_TOKEN }
```

Ref: [Deploy PR documentation](#)

/ Hk tag repository version

Tag repository version

```
- uses: actions/checkout@v6
  with:
    fetch-depth: 0
    fetch-tags: true
- name: Update latest vX and vX.Y tag
  uses: ansys/actions/hk-tag-repository-version@v10.3
  with:
    gh-token: ${ secrets.GITHUB_TOKEN }
    release-version: ${ github.event.release.tag_name }
```

Ref: [Tag repository version](#)

Tag repository version

```
- uses: actions/checkout@v6
  with:
    fetch-depth: 0
    fetch-tags: true
- env:
  SEMANTIC_VERSION: ${ github.event.release.tag_name }
  id: major_minor_version
  name: vX.Y version from tag
  run: 'MAJOR_VERSION=$(echo ${SEMANTIC_VERSION} | cut -d
    -f 1)

  MINOR_VERSION=$(echo ${SEMANTIC_VERSION} | cut -d
    2)

  echo "MAJOR_MINOR_VERSION=v${MAJOR_VERSION}.${MINOR
    }"

  shell: bash
- name: Update latest vX and vX.Y tag
  uses: ansys/actions/hk-tag-repository-version@v10.3
  with:
    gh-token: ${ secrets.GITHUB_TOKEN }
    latest-branch-name: release/${ steps.major_minor_v
    }
```

Ref: [Tag repository version](#)

/ Check licenses

Check library dependencies ship with valid licenses

```
- name: Validate third party licenses
  uses: ansys/actions/check-licenses@v10.3
  with:
    python-version: ${ env.MAIN_PYTHON_VERSION }
```

Ref: [Check library dependencies ship with valid licenses](#)

/ Release pypi private

Release to private pypi

```
- name: Release to the private PyPI repository
  uses: ansys/actions/release-pypi-private@v10.3
  with:
    library-name: ansys-<product>-<library>
    twine-token: ${{ secrets.PYANSYS_PYPI_PRIVATE_PAT }}
    twine-username: __token__
```

Ref: [Release to private PyPI](#)

/ Check pr title

Check pull-request title follows conventional commits

```
- uses: ansys/actions/check-pr-title@v10.3
  with:
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Check pull-request title follows conventional commits](#)

/ Doc changelog

Create changelog fragment

```
- uses: ansys/actions/doc-change@v10.3
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: [Create changelog fragment](#)

/ Docker style

Docker style

```
- uses: ansys/actions/docker-style@v10.3
  with:
    directory: docker
    error-level: 1
    recursive: true
```

Ref: [Docker style](#)

/ Doc deploy dev

Deploy developers documentation

```
- name: Deploy the latest documentation
  uses: ansys/actions/doc-deploy-dev@v10.3
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: <library>.docs.pyansys.com
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Deploy developers documentation](#)

/ Release pypi test

Release to test pypi

```
- name: Release to the test PyPI repository
  uses: ansys/actions/release-pypi-test@v10.3
  with:
    library-name: ansys-<product>-<library>
    twine-token: ${{ secrets.PYANSYS_PYPI_TEST_PAT }}
    twine-username: __token__
```

Ref: [Release to test PyPI](#)

/ Build wheelhouse

Build wheelhouse for latest python versions

```
- name: Build a wheelhouse of the Python library
  uses: ansys/actions/build-wheelhouse@v10.3
  with:
    library-name: <ansys-product-library>
    operating-system: ${{ matrix.os }}
    python-version: ${{ matrix.python-version }}
```

Ref: [Build wheelhouse for latest Python versions](#)

/ Hk package clean untagged

Perform untagged versions cleanup

```
- name: Perform untagged versions cleanup
  uses: ansys/actions/hk-package-clean-untagged@v10.3
  with:
    package-name: mypackage
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Perform untagged versions cleanup](#)

/ Code style

Running code style checks

```
- name: Run PyAnsys code style checks
  uses: ansys/actions/code-style@v10.3
```

Ref: [Running code style checks](#)

/ Doc style

Running documentation style checks

```
- name: Running documentation style checks
  uses: ansys/actions/doc-style@v10.3
  with:
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Running documentation style checks](#)

/ Check vulnerabilities

Check library vulnerabilities

```
- uses: ansys/actions/check-vulnerabilities@v10.3
  with:
    dev-mode: ${{ github.ref != 'refs/heads/main' }}
    python-package-name: ansys-<product>-<library>
    python-version: ${{ env.MAIN_PYTHON_VERSION }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: [Check library vulnerabilities](#)

/ Hk automerge prs

Automerging dependabot and pre-commit.ci prs

```
- name: Automerging dependabot and pre-commit.ci PRs
  uses: ansys/actions/hk-automerge-prs@v10.3
  with:
    approver: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    approver-token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: [Automerging dependabot and pre-commit.ci PRs](#)

/ Release github

Release to github

```
- name: Release to GitHub
  uses: ansys/actions/release-github@v10.3
  with:
    library-name: ansys-<product>-<library>
```

Ref: [Release to GitHub](#)

/ Doc deploy stable

Deploy stable documentation

```
- name: Deploy the stable documentation
  uses: ansys/actions/doc-deploy-stable@v10.3
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: <library>.docs.pyansys.com
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Deploy stable documentation](#)

/ Build ci wheels

Build a c-extension library

```
- name: Build a C-extension library wheel artifacts
  uses: ansys/actions/build-ci-library@v10.3
```

Ref: [Build a C-extension library](#)

/ Build library

Build library basic example

```
- name: Build library source and wheel artifacts
  uses: ansys/actions/build-library@v10.3
  with:
    library-name: ansys-<product>-<library>
```

Ref: [Build library basic example](#)

/ Release pypi public

Release to public pypi

```
- name: Release to the public PyPI repository
  uses: ansys/actions/release-pypi-public@v10.3
  with:
    library-name: ansys-<product>-<library>
    twine-token: ${{ secrets.PYPI_TOKEN }}
    twine-username: __token__
```

Ref: [Release to public PyPI](#)

/ Hk package clean except

Perform versions cleanup - except certain tags

```
- name: Perform versions cleanup - except certain tags
  uses: ansys/actions/hk-package-clean-except@v10.3
  with:
    package-name: mypackage
    tags-kept: latest, latest-unstable
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: [Perform versions cleanup - except certain tags](#)

/ Doc deploy changelog

Update changelog for new tag

```
- uses: ansys/actions/doc-deploy-changelog@v10.3
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: [Update CHANGELOG for new tag](#)